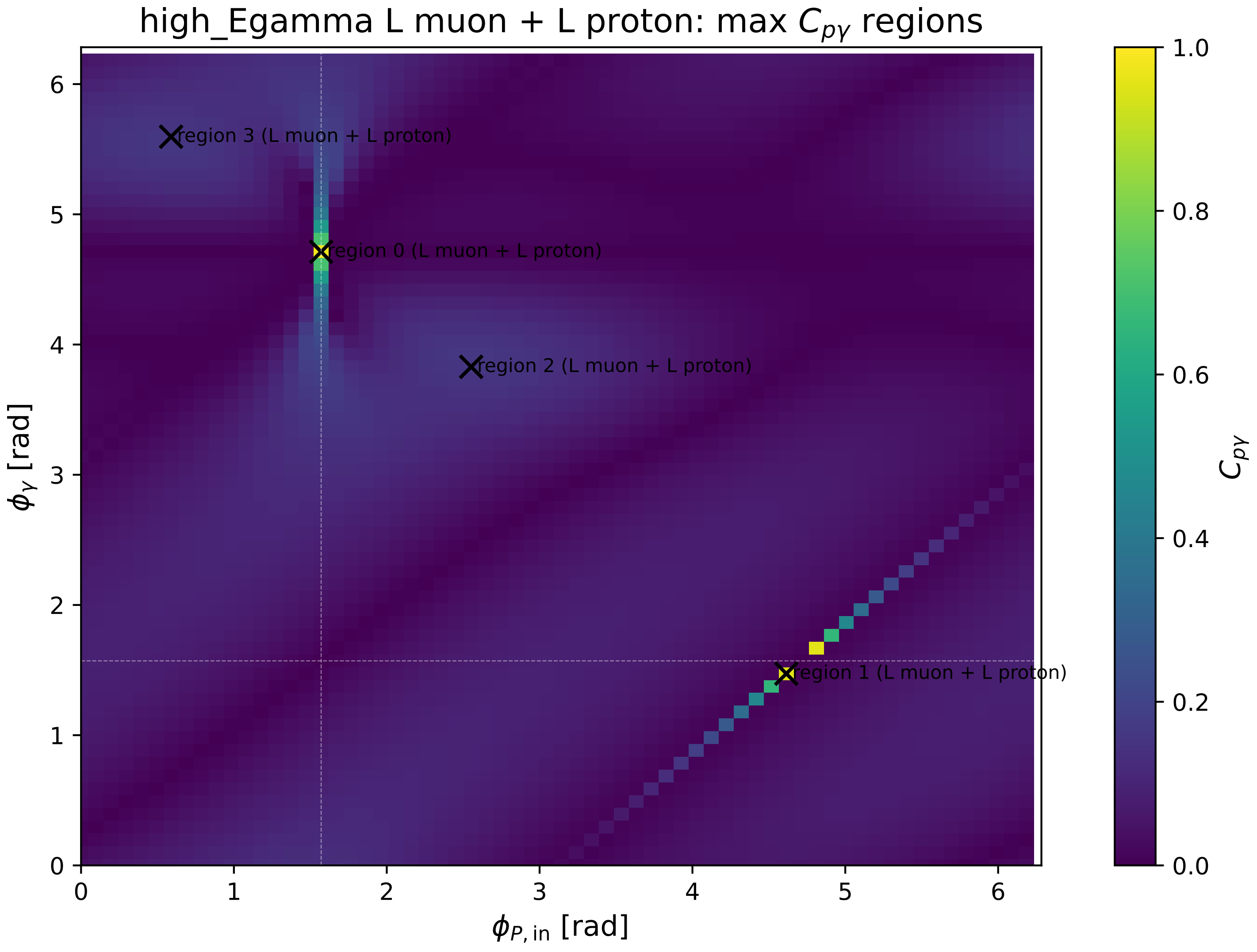
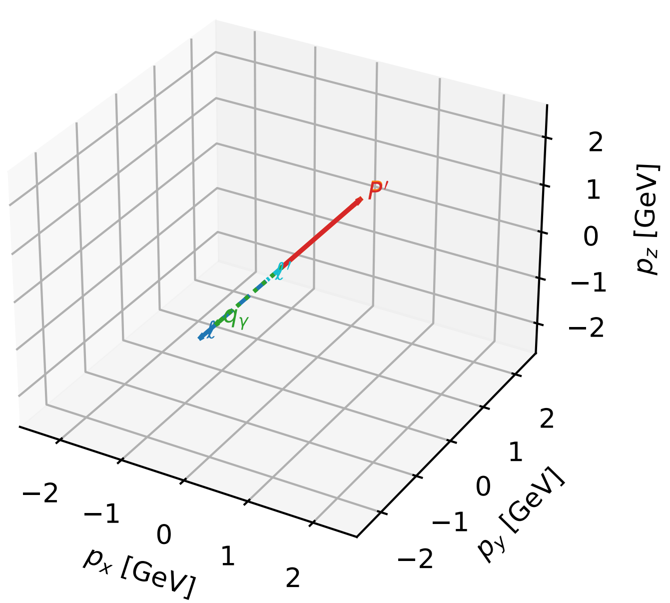


high_Egamma L muon + L proton: max $C_{p\gamma}$ regions
region 3 (L muon + L proton)
region 0 (L muon + L proton)
region 2 (L muon + L proton)
region 1 (L muon + L proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{p\gamma}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

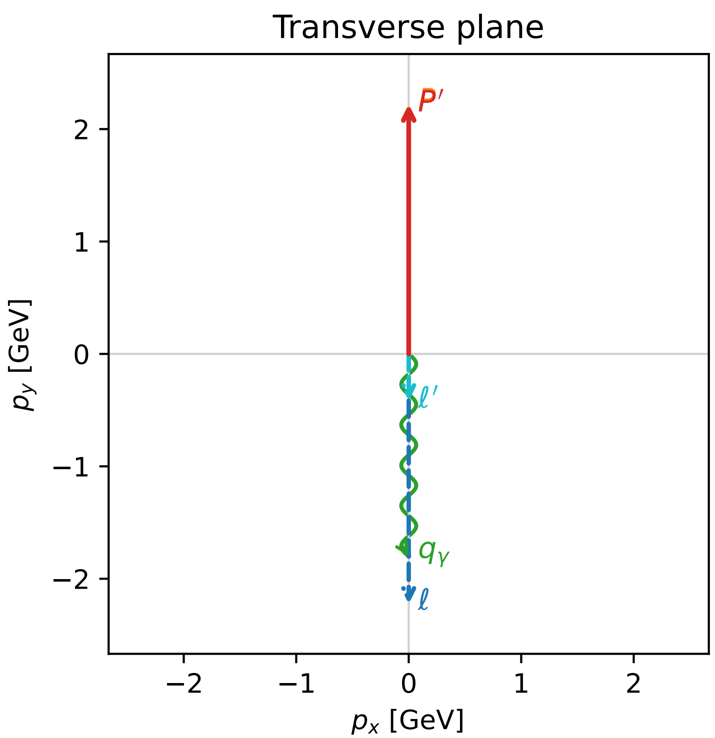
L muon + L proton characteristic max $C_{p\gamma}$ configuration

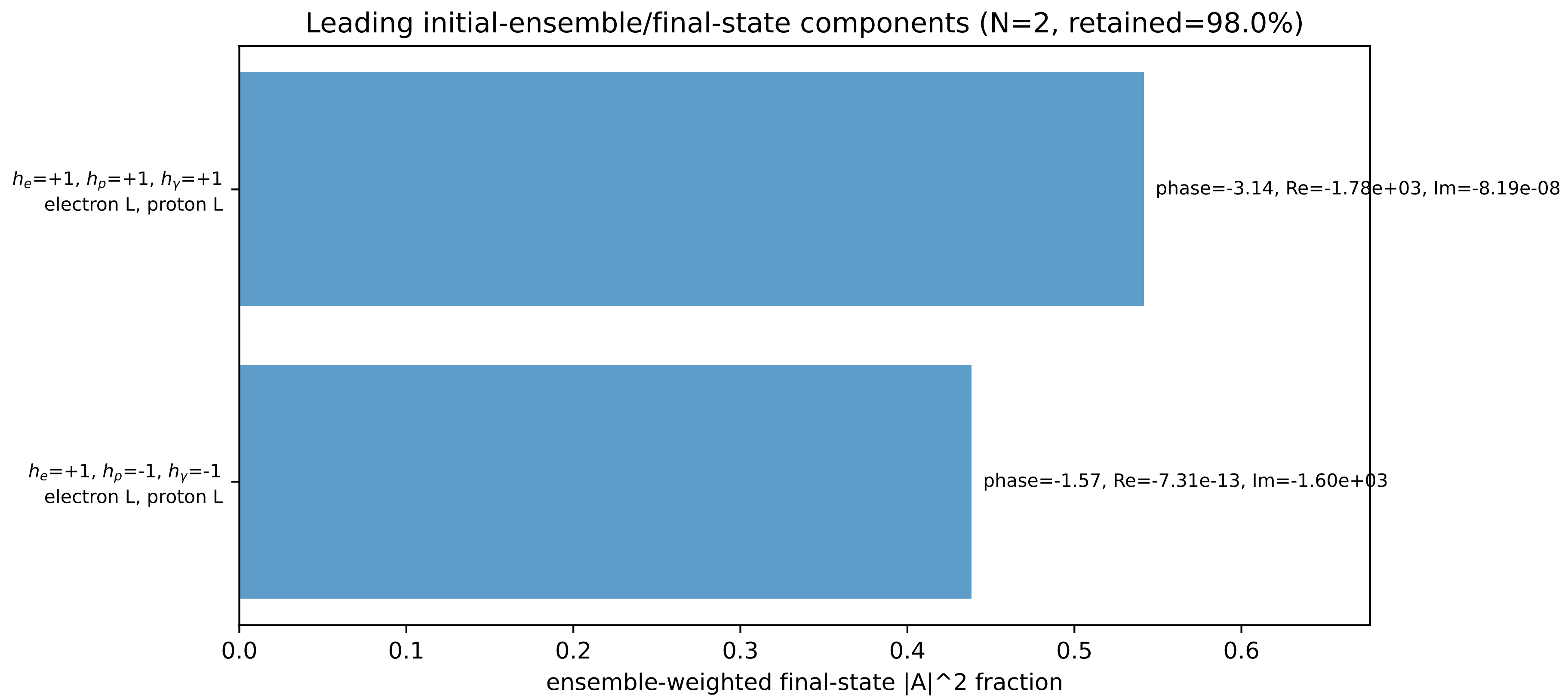


c_p_gamma_high_egamma_ll_region_0 C_{p\gamma}=0.974551
region: high_Egamma_LL_region_0 final-pair delta_xy=3.14159 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.21757
θ_{in}=1.57079, ϕ_ℓ=4.71239, ϕ_P=1.5708
E_γ=1.8, ϕ_γ=4.71239
Q^2=0.0383473, x_B=0.00229767, t=-4.65453e-06, W^2=17.5311, y=0.809201
$\theta(\ell', \gamma)$=0 deg

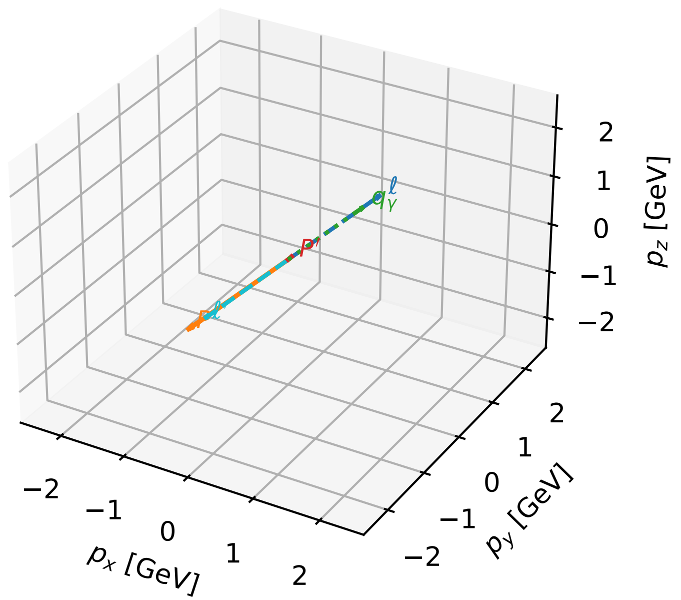
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -0.0000 p_y= -2.2231 p_z= -0.0000
P E= 2.4129 p_x= 0.0000 p_y= 2.2231 p_z= 0.0000
ℓ' E= 0.4307 p_x= 0.0000 p_y= -0.4176 p_z= 0.0000
P' E= 2.4078 p_x= 0.0000 p_y= 2.2176 p_z= 0.0000
q_γ E= 1.8000 p_x= -0.0000 p_y= -1.8000 p_z= 0.0000





L muon + L proton characteristic max $C_{p\gamma}$ configuration

3D momenta

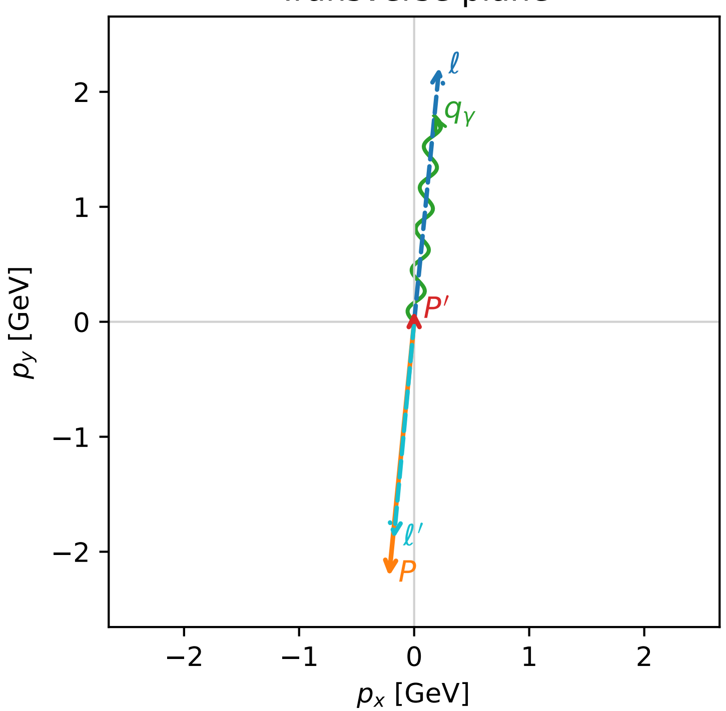



```
c_p_gamma_high_egamma_ll_region_1  C_{p\gamma}=0.95678
region: high_Egamma_LL_region_1  final-pair delta_xy=0.0981748 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
```
incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

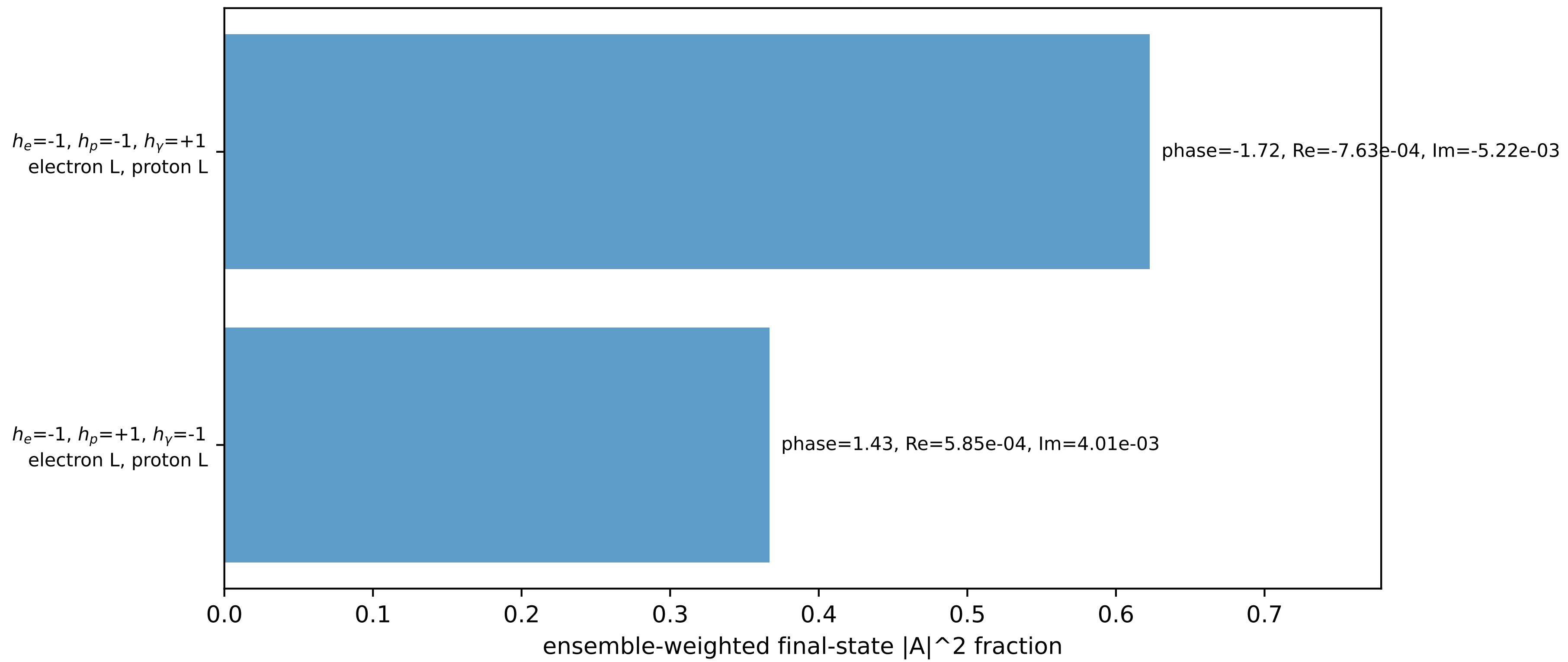
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=0.0933634
θ_{in}=1.57079, ϕ_ℓ=1.47262, ϕ_P=4.61421
E_γ=1.8, ϕ_γ=1.47262
Q^2=16.833, x_B=0.84622, t=-3.20239, W^2=3.93884, y=0.964469
$\theta(\ell', \gamma)$=179.723 deg

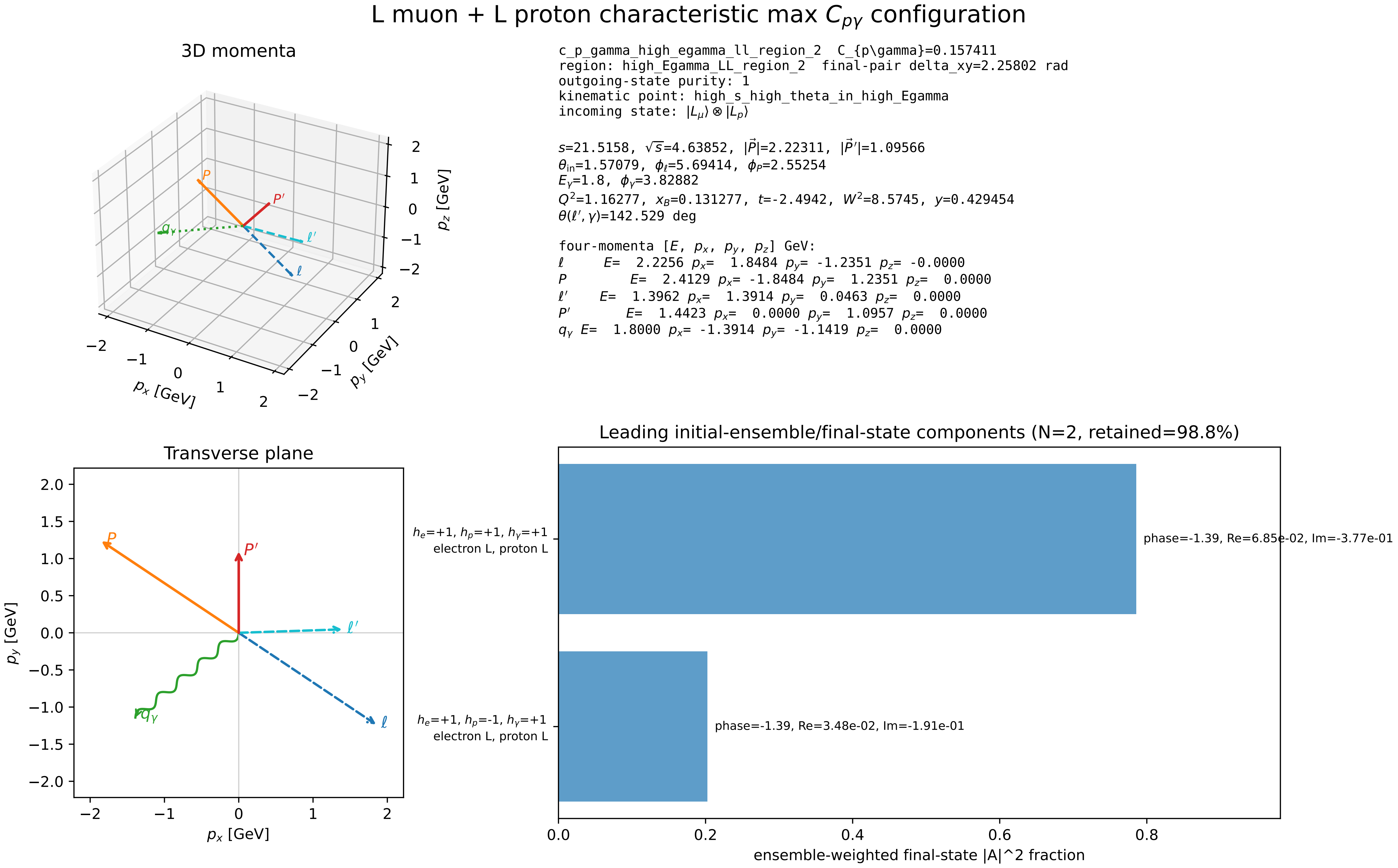
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 0.2179 p_y= 2.2124 p_z= -0.0000
P E= 2.4129 p_x= -0.2179 p_y= -2.2124 p_z= 0.0000
ℓ' E= 1.8959 p_x= -0.1764 p_y= -1.8847 p_z= 0.0000
P' E= 0.9426 p_x= 0.0000 p_y= 0.0934 p_z= 0.0000
q_γ E= 1.8000 p_x= 0.1764 p_y= 1.7913 p_z= 0.0000

Transverse plane

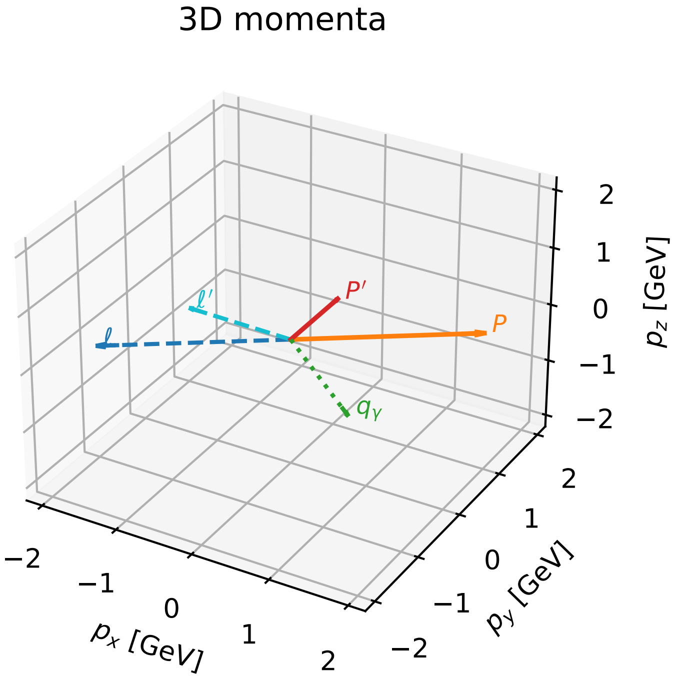


Leading initial-ensemble/final-state components (N=2, retained=99.0%)




L muon + L proton characteristic max $C_{p\gamma}$ configuration
3D momenta
P
P'
q_γ
ℓ'
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
c_p_gamma_high_egamma_ll_region_2 C_{p\gamma}=0.157411
region: high_Egamma_LL_region_2 final-pair delta_xy=2.25802 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $|L_\mu\rangle \otimes |L_p\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.09566
θ_{in}=1.57079, ϕ_ℓ=5.69414, ϕ_P=2.55254
E_γ=1.8, ϕ_γ=3.82882
Q^2=1.16277, x_B=0.131277, t=-2.4942, W^2=8.5745, y=0.429454
$\theta(\ell', \gamma)$=142.529 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 1.8484 p_y= -1.2351 p_z= -0.0000
P E= 2.4129 p_x= -1.8484 p_y= 1.2351 p_z= 0.0000
ℓ' E= 1.3962 p_x= 1.3914 p_y= 0.0463 p_z= 0.0000
P' E= 1.4423 p_x= 0.0000 p_y= 1.0957 p_z= 0.0000
q_γ E= 1.8000 p_x= -1.3914 p_y= -1.1419 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=2, retained=98.8%)
h_e=+1, h_p=+1, h_γ=+1
electron L, proton L
phase=-1.39, Re=6.85e-02, Im=-3.77e-01
h_e=+1, h_p=-1, h_γ=+1
electron L, proton L
phase=-1.39, Re=3.48e-02, Im=-1.91e-01
ensemble-weighted final-state |A|^2 fraction

L muon + L proton characteristic max $C_{p\gamma}$ configuration



c_p_gamma_high_egamma_ll_region_3 C_{p\gamma}=0.157411
region: high_Egamma_LL_region_3 final-pair delta_xy=2.25802 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.09566
$\theta_{\rm in}$=1.57079, ϕ_ℓ=3.73064, ϕ_P=0.589049
E_γ=1.8, ϕ_γ=5.59596
Q^2=1.16277, x_B=0.131277, t=-2.4942, W^2=8.5745, y=0.429454
$\theta(\ell',\gamma)$=142.529 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -1.8484 p_y= -1.2351 p_z= -0.0000
P E= 2.4129 p_x= 1.8484 p_y= 1.2351 p_z= 0.0000
ℓ' E= 1.3962 p_x= -1.3914 p_y= 0.0463 p_z= 0.0000
P' E= 1.4423 p_x= 0.0000 p_y= 1.0957 p_z= 0.0000
q_γ E= 1.8000 p_x= 1.3914 p_y= -1.1419 p_z= 0.0000